Revista de Informática Teórica e Aplicada - RITA - ISSN 2175-2745
Vol. XX, Num. XX (YYYY) XX-XX

RESEARCH ARTICLE

Statistical Forensics of the 2026 Brazilian Presidential Election at the Polling-Section Level: a CRISP-DM Application to the Electoral Court's Open Data

Forense Estatística da Eleição Presidencial Brasileira de 2026 por Seção Eleitoral: uma Aplicação da Metodologia CRISP-DM aos Dados Abertos do TSE

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Abstract: Brazil publishes the vote count and the files of every voting machine, which allows anyone to check whether the 2026 presidential result is consistent with what the machines recorded. We audited both rounds (4 and 25 October; 499,206 and 499,205 polling sections) with the Superior Electoral Court (TSE) open data, in CRISP-DM phases, using eight statistical forensic tests (T1–T8), validated beforehand on synthetic data with inserted frauds, and eleven further tests (N1–N12, except N3) that include the machines' own files. Recounting every vote in the digital vote records (RDV) reproduced the tally sheet (BU) in all 3,030,617 section–office comparisons, and the BUs reproduced the official totals without a single vote of difference; the SHA-512 hashes and the hardware signatures of all 998,414 section files were valid. Machine logs showed no votes before opening or after closing and at most four votes per minute. Digit, integer-percentage, local-homogeneity and fingerprint tests were within the expected range; the first round explained 93% of the within-municipality variance of the runoff; the machine model had no effect within the same polling place; turnout change between rounds was linked to the unexplained vote shift by a gradient of at most 0.22 percentage points whose sign varied across states; and sections that gained voters shifted symmetrically between finalists (88 vs. 70; $p = 0.18$). A random forest trained on synthetic frauds flagged 2 real sections, against 499 expected from its false-positive rate. Remaining outliers (at most about 0.2% of sections, depending on the test) include a disproportionate number of indigenous villages, prisons and sections abroad, and municipalities with bimodal vote distributions all contain indigenous lands. We list the sections for manual verification, report the limits of each test and release the code. Because the digital vote record and the tally sheet are produced by the same machine software, these checks establish consistency and post-recording integrity, not that each vote was recorded as cast.
Keywords: Election forensics — CRISP-DM — Open government data — Electronic voting — Benford's law — Electoral integrity

Resumo: O Brasil publica a contagem de votos e os arquivos de cada urna eletrônica, o que permite conferir se o resultado presidencial de 2026 é consistente com o que as urnas registraram. Auditamos os dois turnos (4 e 25 de outubro; 499.206 e 499.205 seções) com os dados abertos do Tribunal Superior Eleitoral (TSE), seguindo as fases do CRISP-DM, com oito testes forenses estatísticos (T1–T8), validados antes em dados sintéticos com fraudes inseridas, e onze testes adicionais (N1–N12, exceto N3), que incluem os arquivos das próprias urnas. A recontagem de todos os votos do Registro Digital do Voto (RDV) reproduziu o Boletim de Urna (BU) nas 3.030.617 comparações seção–cargo, e os BUs reproduziram os totais oficiais sem um único voto de diferença; os *hashes* SHA-512 e as assinaturas de *hardware* dos arquivos das 998.414 seções foram válidos. Os *logs* não mostraram votos antes da abertura nem depois do encerramento, e no máximo quatro votos por minuto. Os testes de dígitos, percentuais inteiros, homogeneidade local e impressão digital ficaram dentro do esperado; o 1º turno explicou 93% da variância do 2º dentro do município; o modelo da urna não teve efeito dentro do mesmo local de votação; a variação de comparecimento entre turnos se associou à mudança não explicada de voto por um gradiente de no máximo 0,22 ponto percentual, com sinal variável entre UFs; e as seções que ganharam eleitores se deslocaram de forma simétrica entre os finalistas (88 contra 70; $p = 0,18$). Uma floresta aleatória treinada com fraudes sintéticas sinalizou 2 seções reais, contra 499 esperadas pela sua taxa de falso positivo. Os casos extremos restantes (no máximo cerca de 0,2% das seções, conforme o teste) incluem número desproporcional de aldeias indígenas, presídios e seções no exterior, e todos os municípios com distribuição bimodal de votos têm terras indígenas. Listamos as seções para conferência, relatamos os limites de cada teste e publicamos o código. Como o Registro Digital do Voto e o Boletim de Urna são produzidos pelo mesmo software da urna, essas verificações atestam consistência e integridade após a gravação, e não que cada voto foi gravado como digitado.
Palavras-Chave: Forense eleitoral — CRISP-DM — Dados abertos — Urna eletrônica — Lei de Benford — Integridade eleitoral

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1. Introduction

Brazil publishes the result of every polling section (*seção eleitoral*), so anyone can reproduce the national tally and test whether the counts behave as an unmanipulated count should. For the 2026 general election the Superior Electoral Court (TSE) released, besides the tally sheet (*Boletim de Urna*, BU), the digital vote record (*Registro Digital do Voto*, RDV) and the log of each voting machine, as well as consolidated files in its Open Data Portal [1].

Election forensics is the branch of statistics that searches such data for the marks that manipulation tends to leave: invented numbers, rounded percentages, coordinated inflation of turnout in some sections, or swings that only a few sections show [2, 3]. In Brazil, Figueiredo Filho, Silva and Carvalho [4] applied five such tests to the 2018 presidential election, and an election-fingerprint analysis of the 2022 runoff found no statistical sign of ballot stuffing or voter rigging [5].

This article applies and extends that line of work to the 2026 presidential election. Four research questions guide the study: **RQ1** — do section-level counts follow the regularities expected from an unmanipulated tally?; **RQ2** — where there are deviations, are they systematic or isolated, and do they have a plausible demographic explanation?; and **RQ3** — are the changes in turnout and in vote profile between the first round and the runoff consistent with each other?; and **RQ4** — does the published result match what the voting machines recorded and signed?

Contributions (i) a reproducible CRISP-DM pipeline, in R and Python, that validates the statistical tests on synthetic data with known inserted frauds before using them on real data; (ii) the use of *local* references — the other sections of the same polling place or municipality — instead of national medians, which removes the regional false alarms we found in a first version; (iii) a new cross-round test (T8) that relates the change in turnout of each section to the part of its vote change that the first-round composition does not explain, with a symmetry test between the two finalists; (iv) a full recount of every digital vote record of both rounds, carried out with code independent of the electoral court's tallying system, with verification of the hashes and hardware signatures of every machine file, and an analysis of the timing of votes in the machine logs; (v) seven further statistical tests (machine model, electorate profile, 2022–2026 panel, round-number spikes, spatial clustering, machine learning and local multimodality); and (vi) lists of flagged sections, classified by the name of the polling place, ready for manual verification.

The study is statistical and exploratory. An unusual pattern indicates where to look; it does not prove irregularity. Likewise, the absence of a pattern does not prove that no irregularity occurred.

2. Theoretical Background

Each forensic test starts from a regularity that honest counts tend to follow and that counts invented or altered by people tend to violate. Table 1 summarizes the tests used here and the main known limitation of each.

The first-digit Benford test was not used. Sections have bounded size, roughly between 100 and 400 voters, and in that range the law does not hold even for clean data [9].

Two statistical cautions run through the whole study. First, with hundreds of thousands of sections any tiny deviation yields $p < 0.001$, so effect size matters more than the p-value. Second, when testing section by section, the false discovery rate must be controlled [16] and the over-dispersion between sections must be corrected; outliers are measured with robust z-scores based on the median and the median absolute deviation [17]. Section 3 gives the formal definitions.

3. Mathematical and Statistical Foundations

This section states the null models, statistics and decision rules used by each test, in the form in which they are implemented in the scripts of Table 4. For section s we write n_s for the number of valid votes (or of votes for the two finalists, when stated), y_s for the votes of a candidate, $\hat{p}_s = y_s/n_s$ for the observed share and π_s for its expected value under the null hypothesis. $m(s)$ denotes the municipality and $\ell(s)$ the polling place of the section.

3.1 Binomial reference and over-dispersion

The common starting point is that, given the share π_s expected for its electorate, the count of a section is binomial,

$$y_s \mid \pi_s \sim \mathrm{Bin}(n_s, \pi_s), \tag{1}$$

with $\mathrm{E}(y_s) = n_s\pi_s$ and $\mathrm{Var}(y_s) = n_s\pi_s(1-\pi_s)$. Real sections vary more than (1) allows, because voters are not assigned to sections at random. Two corrections are used. In the quasi-binomial approach [18] the variance is multiplied by a dispersion factor estimated from the Pearson residuals,

$$\hat{\varphi} = \max\Big\{1,\ \frac{1}{N}\sum_{s}\frac{(y_s - n_s\bar{\pi}_{\ell(s)})^2}{n_s\bar{\pi}_{\ell(s)}(1-\bar{\pi}_{\ell(s)})}\Big\}, \tag{2}$$

where $\bar{\pi}_\ell$ is the pooled share of the polling place. In the beta-binomial model [19] π_s itself varies around its mean with intra-section correlation ρ, so that

$$\mathrm{Var}(y_s) = n_s\pi_s(1-\pi_s)\big[1 + (n_s-1)\rho\big], \tag{3}$$

and ρ is estimated by the method of moments,

$$\hat{\rho} = \max\Big\{0,\ \frac{\sum_s (r_s^2 - 1)(n_s - 1)}{\sum_s (n_s - 1)^2}\Big\}, \tag{4}$$

with r_s the Pearson residual under (1). Models on the share scale use the empirical logit $\lambda_s = \ln\{(y_s + \frac{1}{2})/(n_s - y_s + \frac{1}{2})\}$, which is finite at 0% and 100% [20].

Table 1. Forensic tests used in this study

Test	Expected regularity	Known limitation	Reference
T1 Zeros and 100%	Few unanimous sections where the candidate has intermediate support	Isolated communities are genuinely unanimous	[2]
T2 Last digit	Digits 0–9 with 10% each in counts with two or more digits	Biased when there are few votes per section	[6, 7]
T3 Second digit (2BL)	Benford distribution of the second digit, mean 4.187	Rejects under normal strategic voting	[8, 9, 10]
T4 Integer percentages	Frequency of integer percentages equal to binomial simulations	Simple fractions create spikes in small sections	[11, 12]
T5 Local homogeneity	Sections of the same polling place are similar	Voters are not randomly assigned to sections	This study (quasi-binomial)
T6 Cross-round consistency	Runoff share predicted by the section's own first round; stable turnout	Transit and transferred voters change the electorate	This study; [13]
T7 Fingerprint	No smear of turnout and leader share toward 100%/100%	Needs eligible voters per section	[3, 14]
T8 Turnout × profile	No link between extra turnout and unexplained shift	Ecological regression assumptions	This study; [15]

3.2 Robust scores and multiple testing

Outliers are measured with the robust z-score

$$z_s = \frac{x_s - \operatorname{med}(x)}{1.4826\,\operatorname{med}\left|x - \operatorname{med}(x)\right|}, \tag{5}$$

where the constant makes the scaled median absolute deviation a consistent estimator of the standard deviation under normality; median and MAD have the highest possible breakdown point, so a few extreme sections do not mask themselves [21, 17]. Thresholds of $|z| \geq 4$ or 6 are therefore conservative even when x has heavier tails than the normal.

When m hypotheses are tested together, the Benjamini–Hochberg procedure [16] sorts the p-values $p_{(1)} \leq \cdots \leq p_{(m)}$ and rejects the k^* smallest, with

$$k^* = \max\{k : p_{(k)} \leq kq/m\}, \tag{6}$$

which keeps the expected proportion of false discoveries below q. For a small family of global tests (eight digit tests, six series of N9) we use the Bonferroni bound α/m [22]. All Monte Carlo and permutation p-values are computed as

$$p = \frac{1 + \#\{b : T^{(b)} \geq T_{\text{obs}}\}}{B+1}, \tag{7}$$

which is exact under the null and never zero [23]; its minimum, $1/(B+1)$, limits the evidence a test with B replicates can provide.

3.3 Count regularities (T1–T5, N9)

T1 With π_s taken as the candidate's share in the *other* sections of the same polling place (or of the municipality, if the place has a single section), the expected numbers of sections with no votes and with all votes are $E_0 = \sum_s (1-\pi_s)^{n_s}$ and $E_{100} = \sum_s \pi_s^{n_s}$. Being a sum of many small, nearly independent probabilities, the observed count is compared with a Poisson distribution, $p = \Pr\{\text{Pois}(E) \geq O\}$.

T2 and T3 Under Benford's law the second significant digit d has probability [24]

$$\Pr(D_2 = d) = \sum_{k=1}^{9} \log_{10}\Big(1 + \frac{1}{10k+d}\Big), \qquad d = 0, \ldots, 9, \tag{8}$$

and, for counts of two or more digits, the last digit is approximately uniform. Observed frequencies O_d are compared with expected frequencies E_d by Pearson's statistic [25]

$$X^2 = \sum_{d=0}^{9} \frac{(O_d - E_d)^2}{E_d} \overset{H_0}{\sim} \chi^2_9. \tag{9}$$

Because bounded section sizes distort both references, E_d is also computed by simulation: each count is redrawn from (1) with $\pi_s = \hat{p}_s$, which preserves the size distribution of the sections.

T4 and N9 A count has an integer percentage when $0 < y_s < n_s$ and $100\,y_s \equiv 0 \pmod{n_s}$; N9 counts sections whose share lies within 0.05 percentage points of a multiple of 5%. In both cases the statistic T is the number of such sections and its null distribution is obtained by redrawing every section from (1) with $\pi_s = \hat{p}_s$, $B = 100$ times in T4 and $B = 500$ times in N9, which is the resampled-kernel idea of Rozenas [12]: spikes produced by small denominators are reproduced, spikes produced by people are not. The p-value follows (7).

T5 Each section is compared with the rest of its polling place by the two-proportion statistic

$$z_s = \frac{\hat{p}_s - \hat{p}_{-s}}{\sqrt{\bar{\pi}_\ell(1-\bar{\pi}_\ell)\big(n_s^{-1} + n_{-s}^{-1}\big)}}, \qquad \tilde{z}_s = z_s/\sqrt{\hat{\varphi}}, \tag{10}$$

where $-s$ denotes the other sections of $\ell(s)$ and $\hat{\varphi}$ is (2). A section is flagged when $\tilde{z}_s$ is significant after (6) with $q = 0.001$ and $|\hat{p}_s - \hat{p}_{-s}| \geq 20$ percentage points.

3.4 Cross-round and turnout models (T6–T8)

T6 The runoff share of a finalist is regressed on the first-round composition of the same section, $\mathbf{c}_s = (c_{s1}, \dots, c_{sK})$, the shares of every candidate and of blank and null votes (one category omitted, since they sum to one), with municipality fixed effects removed by the weighted within transformation (a double dot denotes the deviation from the municipal mean weighted by n_s, the regression weight):

$$\ddot{p}_s^{(2)} = \sum_{k=1}^{K-1} \beta_k \, \ddot{c}_{sk} + e_s, \tag{11}$$

The R^2 of (11) is the within-municipality R^2 reported in Section 6; the scaled residuals $e_s\sqrt{n_s}$ are converted to robust scores (5). Turnout change is $\Delta C_s = \ln(V_{2,s}/V_{1,s})$, centred on the municipality.

T7 The fingerprint is the two-dimensional histogram of turnout $t_s = V_s/A_s$ (A_s eligible voters) and leader share. In the model of Klimek et al. [3], ballot stuffing moves a fraction of the abstentions and of the opponent's votes to the leader in a subset of units, which produces a second mode near $(t, \hat{p}) = (1, 1)$ and a positive association between t_s and $\hat{p}_s$ within regions; both are checked.

T8 Goodman's ecological regression [15] estimates, by state and without intercept, how first-round votes transfer to each finalist:

$$V_{2,s}^{A} = \sum_{j \in \{A,B,O,BN\}} \gamma_j^{A} V_{1,s}^{j} + u_s, \tag{12}$$

and analogously for B, where O are the other candidates and BN blank and null votes. The predicted share is $\tilde{p}_{2,s} = \hat{V}_{2,s}^{A}/(\hat{V}_{2,s}^{A} + \hat{V}_{2,s}^{B})$ and the unexplained shift is $\Delta P_s = 100(p_{2,s} - \tilde{p}_{2,s})$. The transfer coefficients γ are valid proportions only if transfer rates are constant within the state; values outside $[0, 1]$ signal that this assumption fails. The association between ΔC and ΔP is measured by Spearman's rank correlation [26], $\rho_S = 1 - 6\sum_i d_i^2/\{N(N^2-1)\}$ with d_i the rank differences, and by decile means with normal 95% intervals. Among the M sections with extreme $\Delta C > 0$ and ΔP, the number X that shifted toward the first finalist is tested against $X \sim \mathrm{Bin}(M, \tfrac{1}{2})$, the distribution expected if neither side benefited.

3.5 Fixed effects and clustered errors (N4–N8)

N4–N7 estimate linear models of the form $y_s = \mathbf{x}_s^{\top}\beta + \alpha_{g(s)} + \varepsilon_s$ with weights w_s and fixed effects α_g for the polling place (N6) or the municipality (N4, N5, N7). The weighted within transformation $\ddot{y}_s = y_s - \sum_{g(i)=g(s)} w_i y_i / \sum_{g(i)=g(s)} w_i$ (and likewise for $\mathbf{x}_s$) removes α_g, and $\hat{\beta} = (\ddot{X}^{\top} W \ddot{X})^{-1} \ddot{X}^{\top} W \ddot{\mathbf{y}}$. Because sections of the same municipality share unobserved shocks, the variance is estimated with the cluster-robust sandwich [27]

$$\widehat{\mathrm{V}}(\hat{\beta}) = c\,\mathbf{H}^{-1}\Big(\sum_{h=1}^{G} \mathbf{g}_h \mathbf{g}_h^{\top}\Big)\mathbf{H}^{-1}, \tag{13}$$

with $\mathbf{H} = \ddot{X}^{\top} W \ddot{X}$ and cluster scores $\mathbf{g}_h = \sum_{s \in h} w_s \ddot{\mathbf{x}}_s \hat{u}_s$ summed over the G municipalities, and small-sample factor $c = G(n-1)/\{(G-1)(n-k)\}$. Two-sided p-values use the normal approximation and the BH correction (6).

In N7 the model is fitted on the empirical logit λ_s; the fitted probability is $\hat{\pi}_s = \mathrm{logit}^{-1}(\mathbf{x}_s^{\top}\hat{\beta} + \hat{\alpha}_{m(s)})$, over-dispersion is estimated from (3), and each section is scored by $z_s = (y_s - n_s\hat{\pi}_s)/\{n_s\hat{\pi}_s(1-\hat{\pi}_s)[1+(n_s-1)\hat{\rho}]\}^{1/2}$. The model is checked by simulation [28, 29]: $R = 200$ replicated data sets $y_s^{(r)} \sim \text{Beta-Bin}(n_s, \hat{\pi}_s, \hat{\rho})$ give the reference distribution of discrepancy statistics $T(\mathbf{y})$ (sections at 0%, above 90% and below 10%, and the within-municipality standard deviation), with two-sided predictive p-value $p = \min\{1, 2\min(\Pr[T(\mathbf{y}^{(r)}) \geq T(\mathbf{y})], \Pr[T(\mathbf{y}^{(r)}) \leq T(\mathbf{y})])\}$. In N8 the change $\delta_s = \lambda_s^{2026} - \lambda_s^{2022}$ of the same section is centred on the municipal median and scored with (5) by state.

3.6 Spatial dependence (N10)

Let z_i be a standardized residual averaged over polling place i and $w_{ij} = 1/k$ if j is one of the $k = 8$ nearest polling places of i in the same state (zero otherwise). Global and local Moran statistics are [30, 31]

$$I = \frac{\sum_i z_i \sum_j w_{ij} z_j}{\sum_i z_i^2}, \qquad I_i = z_i \sum_j w_{ij} z_j. \tag{14}$$

$I > 0$ means that neighbouring places have similar residuals. The local statistic is assessed by conditional permutation: the neighbours of i are replaced $B = 199$ times by places drawn at random, and p_i follows (7), with (6) at $q = 0.01$. Since $p_i \geq 1/(B+1) = 0.005$ and (6) over $m \approx 9 \times 10^4$ places requires $p_{(k)} \leq kq/m$, no place can be significant unless at least $0.005\,m/q \approx 4.6 \times 10^4$ places reach that minimum together; this is why the null result of N10 is uninformative (Section 9).

3.7 Classification and multimodality (N11, N12)

N11 A random forest [32, 33] estimates, from synthetic sections, the probability $\hat{f}(\mathbf{x})$ that a section with features $\mathbf{x}$ was manipulated. Discrimination is measured by the area under the ROC curve, which equals the Mann–Whitney probability that a manipulated section receives a higher score than a clean one [34, 35]:

$$\mathrm{AUC} = \frac{R_1 - n_1(n_1+1)/2}{n_1 n_0}, \tag{15}$$

with R_1 the rank sum of the n_1 manipulated sections among all $n_1 + n_0$ scores. The decision threshold is the $(1-\alpha)$ quantile of the scores of held-out clean sections, $\tau = \hat{F}_0^{-1}(1-\alpha)$ with $\alpha = 0.001$, so that among N clean real sections about αN would be flagged by chance.

N12 Hartigan's dip [36] is the distance between the empirical distribution function F_n of the section shares of a municipality and the closest unimodal distribution,

$$D(F_n) = \inf_{G \in \mathscr{U}} \sup_{x} \big|F_n(x) - G(x)\big|, \tag{16}$$

where $\mathscr{U}$ is the class of unimodal distribution functions. Its p-value is taken from the null distribution under the uniform, the least favourable unimodal case, and the municipalities are corrected with (6) at $q = 0.05$.

3.8 Integrity of the machine files (N1, N2)

N1 is an identity test: for every section and office, the vector of votes recounted from the RDV must equal the vector printed in the BU, and the BU must equal the published CSV. N2 relies on two primitives. SHA-512 [37] maps a file to a 512-bit digest $h = H(F)$; a change in any bit of F changes h unpredictably, so equality of the recomputed digest with the signed one shows that the file is the one that was signed. The signature on the digest list is verified with the public key embedded in the machine certificate. For ECDSA over P-521 [38, 39], with base point G of order q, public key Q, message digest e and signature (r, s), the verifier computes

$$w = s^{-1} \bmod q, \quad (x_1, y_1) = (ew \bmod q)\, G + (rw \bmod q)\, Q, \tag{17}$$

and accepts iff $x_1 \equiv r \pmod q$. For EdDSA [40, 41] on the Edwards curve E-521, $x^2 + y^2 = 1 - 376014\, x^2 y^2$ over $\mathbb{F}_{2^{521}-1}$ [42], a signature (R, S) on message M under public key A is accepted iff

$$[S]B = R + [k]A, \qquad k = H(R \,\|\, A \,\|\, M), \tag{18}$$

with B the base point. Forging either signature without the private key requires solving the elliptic-curve discrete logarithm problem, so a valid signature binds the files to the key of the certificate they carry; it does not, by itself, show that the key belongs to a genuine machine.

4. Related Work

The literature on election forensics can be organized into four strands: digit tests, distributional tests on vote shares and turnout, cross-round or cross-election comparisons, and supervised learning on synthetic frauds. Table 2 compares the studies closest to ours.

Digit tests Mebane [8] proposed the second-digit Benford test (2BL) for precinct counts in United States presidential elections. Pericchi and Torres [10] applied first- and second-digit tests, plus a version restricted to bounded units, to elections in the United States, Puerto Rico and Venezuela; the second-digit law was clearly rejected only in the 2004 Venezuelan recall referendum and only for electronic voting units. Beber and Scacco [6] moved the focus to last digits and digit pairs, exploiting the biases people show when inventing numbers. Deckert, Myagkov and Ordeshook [9] showed that Benford-type tests can fail in both directions, and an experiment by Mack and Stoetzer [7], in which participants were paid to alter tally sheets without being detected, found that last-digit tests do not reliably identify manipulated counts. These results are why we treat T2 and T3 as weak evidence and calibrate them by simulation.

Vote-share and turnout distributions Klimek et al. [3] introduced the turnout–vote *fingerprint* and a model of ballot stuffing and voter rigging; the same group later found fingerprints of ballot stuffing in about 11% of Turkish polling stations in the 2017 referendum [14]. Kobak, Shpilkin and Pshenichnikov [11] showed that falsified Russian results contain an excess of integer percentages, and Rozenas [12] proposed a resampled kernel-density method that separates such spikes from those created by small units. Leemann and Bochsler [43] argued for testing several empirical implications of different forms of fraud together and discussed how to handle the resulting multiple comparisons, a point we follow with the Benjamini–Hochberg procedure.

Cross-round comparisons Closest to our T6 and T8, Klimek, Aykaç and Thurner [13] compared the two rounds of the 2023 Turkish presidential election at the ballot-box level and found that districts with few ballot boxes were more likely to show large vote swings toward the incumbent between rounds. Their test is about the size of the swing in remote units; ours relates the swing that the first round does not explain to the change in turnout and tests whether extreme cases favour one side.

Machine learning on synthetic frauds Cantú and Saiegh [44] trained a classifier on synthetic clean and manipulated elections and applied it to Argentina in the 1930s; Zhang, Alvarez and Levin [45] used random forests trained on synthetic data to flag polling places in Argentina's 2015 election. Cantú [46] used convolutional neural networks on images of about 50,000 tally sheets from Mexico's 1988 election and found blatant alterations in about a third of them, showing the value of checking the physical record and not only the published numbers. Our validation step borrows the idea of synthetic frauds, but uses it to calibrate and debug classical tests rather than to train a classifier.

Brazil Figueiredo Filho, Silva and Carvalho [4] applied five tests (2BL, last-digit mean, frequency of last digits 0 and 5, correlation between vote share and turnout, and resampled kernel density) to the 2018 presidential election at the section and municipality levels and found no evidence of fraud. For the 2022 runoff, Klimek and Thurner [5] analysed 471,983 units and found no fingerprints of ballot stuffing or voter rigging, and Nogueira [47] discussed statistical methods and electoral integrity in the 2022 Brazilian elections. Other Brazilian applications of Benford's law target different objects: campaign donations [48] and Senate elections from 1994 to 2018 with a change of numerical base [49].

Security of the Brazilian voting machine A separate line of work examines the machine itself. Rivest [50] defines a voting system as *software independent* if an undetected change or error in its software cannot cause an undetectable change in the outcome; paperless direct-recording machines such as the Brazilian one are software dependent. In the Public Security Tests of the Electoral Court, Aranha et al. recovered the order

of the votes from the digital vote record, whose shuffling was seeded by the machine clock [51], and later executed arbitrary code on the machine [52]. Van de Graaf [53] summarizes the consequences for non-specialists: the digital vote record is a second copy written by the same software, not a physical record verified by the voter, so it cannot reveal a fault in how votes were captured; and because the poll worker can disable biometric identification, absent voters can in principle be impersonated, which the machine logs may reveal. These works do not test election data, but they define what a data audit such as ours can and cannot show.

Gap The Brazilian studies above analysed one round at a time, used national references, relied only on the published counts and did not report how each test behaves on data with known frauds. This study analyses both rounds, links them section by section and to 2022, uses local references, validates the statistical tests on synthetic frauds before using real data, and complements statistical forensics with the direct check that Cantú [46] showed to be decisive: the comparison of the published result with the primary record — here, the files signed by each voting machine.

5. Methodology

The study followed the six phases of CRISP-DM [54, 55], with the evaluation phase carried out *before* the use of real data (Figure 1). All processing was done in R with the `data.table` package.

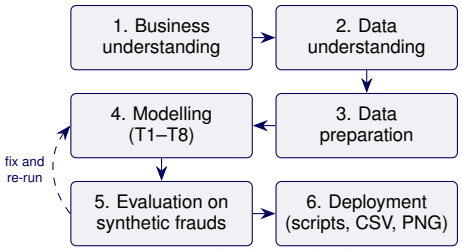


Figure 1. CRISP-DM as applied. Evaluation on synthetic data with known frauds came before the analysis of real data and fed corrections back into modelling.

Business understanding The question is whether section-level counts have regularities compatible with an unmanipulated tally. The success criterion is that each test separates clean data from data with inserted fraud, without false alarms.

Data understanding The sources are the files "President — vote by polling section" (`votacao_secao_2026_BR`) and "Count details by polling section" (`detalhe_votacao_secao_2026`), generated by the TSE on 26 October 2026 [1]. The first has one row per section, round and candidate, with state (UF), municipality, zone, section, polling place and votes; blank votes have code 95 and null votes code 96. The second has eligible voters, turnout and the voting-machine model per section. State files contain only state-level offices. For the further tests we also used, from the same portal, the electorate profile per section (sex, age group, schooling and biometric registration), the coordinates of the polling places [56], the 2022 presidential vote per section [57] and the "files transmitted for tallying": one ZIP per state and round with the BU, RDV, log and signature files of every voting machine [58], up to 25.8 GB per file (São Paulo, runoff).

Data preparation Rows were filtered for the office of President and converted from Latin-1 to UTF-8. The TSE does not publish rows with zero votes, so each section was completed with zeros for all candidates of the round. The section key is UF + municipality + zone + section; valid votes exclude codes 95, 96 and 97. A load check compared both files: the sum of votes equals the reported turnout in *every* section of both rounds (125,275,835 votes in the first round and 128,236,074 in the runoff, for 158,745,502 eligible voters; turnout of 78.9% and 80.8%). The detail file lists 43 further sections abroad (42 in the first round and 43 in the runoff) with eligible voters but zero turnout, and therefore no votes.

Modelling Eight statistical tests, T1 to T8, described in Sections 6 and 7, and eleven further tests, N1 to N12, described in Sections 8 and 9 (N3, the comparison of printed and published tally sheets, needs field samples and was not run). References are local: the other sections of the same polling place or municipality, not the national median. Section-level tests use the Benjamini–Hochberg correction ($q < 0.001$) and robust z-scores [17].

Evaluation We generated 3,559 synthetic sections in the TSE layout, in two versions. One is clean. The other has three inserted frauds: 8 sections with 100% for one candidate, 300 sections with a forced integer percentage and 5 sections with inflated turnout. This step revealed three faults that were fixed before the real analysis: the theoretical digit references raised false alarms; the homogeneity test without dispersion adjustment flagged 75 clean sections; and turnout compared with the national median confused regional variation with anomaly. Table 3 shows the corrected tests.

Deployment The pipeline is a set of numbered scripts (Table 4) that share one configuration file and can be run in sequence by a single command. Each script starts with a load check and writes its results as CSV and PNG files, ready for checking against the tally sheets. The machine files are decoded in Python with the ASN.1 specification published by the TSE [59], reusing the parsers of the open-source DataUrnas-BR project [60]; processing both rounds took 143 minutes on a desktop computer with 15 parallel processes.

6. Results

Both finalists passed every aggregate regularity test. Deviations appear in few sections and, in general, repeat the same pattern in both rounds. We analysed 499,206 sections in the first round and 499,205 in the runoff.

Table 2. Related studies at the polling-unit level

Study	Election(s)	Methods	Main finding
Pericchi and Torres [10]	USA, Puerto Rico, Venezuela	First- and second-digit Benford, restricted version	2BL rejected only for Venezuela 2004 electronic units
Cantú and Saiegh [44]	Argentina 1931–1941	Naive Bayes trained on synthetic data	Historically irregular elections classified as fraudulent
Klimek et al. [3]	Several countries	Turnout–vote fingerprint, fraud model	Fingerprints of ballot stuffing in some countries
Beber and Scacco [6]	Several, incl. Nigeria	Last digits and digit pairs	Human-invented counts leave digit traces
Leemann and Bochsler [43]	Swiss referendum 2011	Several tests, multiple-comparison control	Irregularities in municipalities that destroyed ballots
Kobak et al. [11]	Russia	Integer percentages	Excess of round percentages
Klimek et al. [14]	Turkey 2017–2018	Fingerprint, fraud model	Ballot stuffing in about 11% of stations (2017)
Zhang et al. [45]	Argentina 2015	Random forest on synthetic data	Polling places at risk flagged
Cantú [46]	Mexico 1988	CNN on tally-sheet images	Alterations in about a third of tallies
Figueiredo Filho et al. [4]	Brazil 2018	Five tests (digits, turnout, kernel)	No evidence of fraud
Klimek and Thurner [5]	Brazil 2022, runoff	Fingerprint, ballot stuffing and rigging	No signs of fraud
Klimek et al. [13]	Turkey 2023, two rounds	Fingerprint, cross-round swings	Extreme swings in remote areas
This study	**Brazil 2026, two rounds**	**T1–T8 with synthetic validation; full RDV recount and signature check; N4–N12**	**See Sections 6–9**

Table 3. Validation on synthetic data (3,559 sections)

Test	Clean data	Inserted frauds
T1 Zeros and 100%	no alert	8 of 8 sections detected
T4 Integer %	ratio 0.95 ($p = 0.72$)	ratio 2.86 ($p < 0.01$)
T5 Local homogeneity	0 sections	8 of 8 detected
T6 Cross-round	0 sections	7 of 8 detected
T6 Turnout	0 sections	4 flagged, 3 inserted
T8 Turnout × profile	1 section (lost votes)	4 of 5 inflated, all "more votes, toward beneficiary"

6.1 Sections with 0% and 100% (T1)

There are more unanimous sections than the binomial model predicts, but the excess is small in absolute terms (Table 5).

Three facts point to communities with homogeneous vote rather than to manipulation. The sections are small (median of 81 valid votes in the listed cases) and more than half lie in the North and in Maranhão, led by PA, AM and MA, with others in MG and RS. The same sections are unanimous in both rounds (for instance AM 2690/280, MA 8958/170 and 179, and PA 4111/516 and 519), and some sections are unanimous for the other finalist in both rounds (RS 85839/81, PA 4200/44). The expected value underestimates unanimity because it assumes voters similar to those of neighbouring sections, which does not hold for villages or isolated communities.

6.2 Digit tests (T2 and T3)

The last digit is uniform for both finalists in both rounds (Figure 2). The theoretical Benford reference for the second digit always rejects ($p \approx 0$), but the simulated reference shows deviations of at most 0.12 percentage points (Table 6).

With eight simultaneous tests the Bonferroni threshold is 0.006. Only the second digit of Flávio Bolsonaro in the runoff falls below it, with a practically irrelevant effect. Mebane [8] already notes that 2BL reacts to ordinary strategic behaviour of voters.

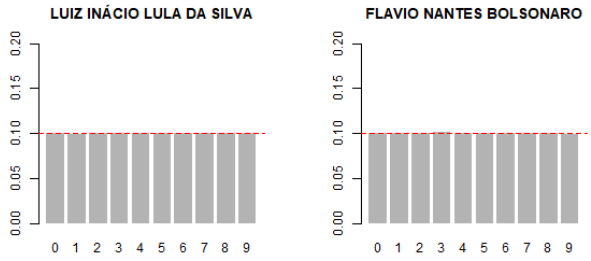


Figure 2. Last digit of the runoff counts for Lula (left) and Flávio Bolsonaro (right); the dashed line is the 10% expected under uniformity.

6.3 Integer percentages (T4)

There is no excess of integer percentages in the runoff: observed/expected ratio 0.995 for both finalists ($p = 0.72$ and 0.69). In the first round the ratios were 1.014 for Lula ($p =$

Table 4. Analysis pipeline

Script	Purpose
`00_config.R`	Paths, helpers, fixed-effect regression, BH
`01_baixar_dados.R`	Download vote and detail files (integrity check)
`02_verificar_carga.R`	Votes = turnout per section; coverage
`03_distribuicao_presidente.R`	Section-level distributions
`04_testes_forenses.R`	T1–T7
`05_cruzamento_comparecimento_perfil.R`	T8
`06_baixar_dados_abertos_extra.R`	Profile, coordinates, 2022 votes
`07_testes_modelo_perfil_painel.R`	N6–N8
`08_testes_padroes_espaciais_ml.R`	N9–N12
`09_arquivos_urna.py`	Decode BU, RDV, log, signatures (N1, N2, N4, N5)
`10_testes_arquivos_urna.R`	Summaries and tests N1, N2, N4, N5
`11_teste_liberacao_biografica.R`	N13, from the files extracted by script 09 (not yet run)
`99_executar_tudo.R`	Runs the whole sequence

Table 5. Unanimous sections (T1)

Round	Event	Obs.	Exp.	p (Poisson)
2nd	100% Lula	106	49.7	2.7×10^{-12}
2nd	100% F. Bolsonaro	4	2.3	0.20
1st	100% Lula	38	18.5	4.5×10^{-5}
1st	0 votes F. Bolsonaro	81	37.4	4.6×10^{-10}

Table 6. Digit tests with simulated references

Rd.	Candidate	Last, p	2nd, p	2nd, max. dev. (p.p.)
1st	Lula	0.48	0.61	0.06
1st	F. Bolsonaro	0.67	0.015	0.09
2nd	Lula	0.76	0.11	0.10
2nd	F. Bolsonaro	0.23	0.0027	0.12

0.059) and 1.009 for Flávio Bolsonaro ($p = 0.18$). Among candidates with enough votes, only Augusto Cury reached the smallest p possible with 100 simulations (0.0099); his excess is about 250 sections, far below the Russian anomalies described by Kobak et al. [11]. The small peaks near 25%, 33% and 50% (Figure 3) come from simple fractions in small sections and are already included in the simulation.

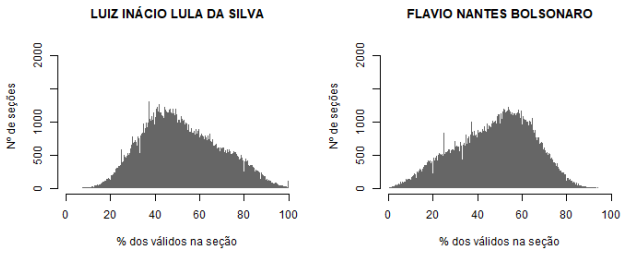


Figure 3. Distribution of the share of valid votes per section in the runoff (x axis: % of valid votes; y axis: number of sections).

6.4 Homogeneity within the polling place (T5)

Sections of the same building are almost as similar as they would be with randomly assigned voters: the dispersion ratio was 1.23–1.24 in the first round and 1.26 in the runoff for the finalists. After the quasi-binomial adjustment and the BH correction, between 67 and 85 sections per candidate and round differ by at least 20 percentage points, 0.014–0.018% of the sections tested. The 46 sections flagged for Lula in both rounds deviate in the same direction in both (e.g. SP 62499/402, MG 41238/336, MA 9610/468), a pattern typical of a different electorate in the section, such as a special section or transferred voters.

6.5 Consistency between rounds (T6)

The first round of each section explains 93% of the variation of the runoff within the municipality ($R^2 = 0.9318$), and the scatter is a narrow band without clusters off the curve (Figure 4).

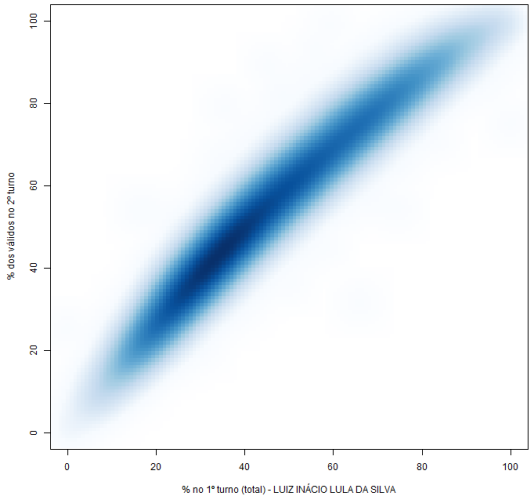


Figure 4. Lula's share in the first round (x axis, % of total) versus his share of valid votes in the runoff (y axis), per section; density plot.

Fourteen sections deviate from the prediction with robust $|z| \geq 6$ (Table 7). Six of them are in indigenous villages or indigenous lands, according to the polling-place name, and one is abroad (Porto, Portugal). Four are in the same polling place in São Paulo (municipality 71072, zone 2), where Flávio Bolsonaro did better than predicted and the sections received 16–25% more votes in the runoff; they may be sections for

voters in transit, since a voter can request transit voting in only one round. This hypothesis was not verified.

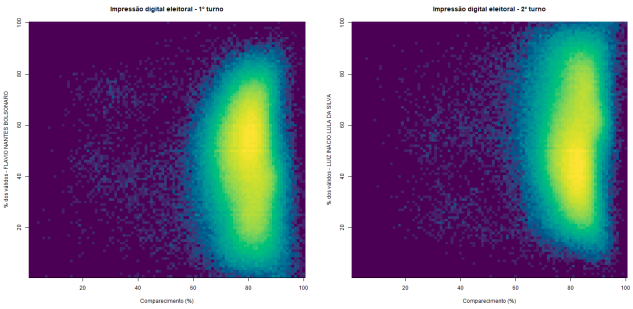
Table 7. Sections flagged by T6 ($|z| \geq 6$)

UF/mun./zone/sec.	Valid	Lula %	z	Polling place (type)
AC 1554/3/160	152	31.6	−14.7	Health post, Buaçú village (indig.)
AM 2097/18/37	155	54.2	13.1	Municipal school
AC 1031/3/95	221	78.7	12.6	Health unit, Maronawa village (indig.)
AM 2399/45/63	169	98.8	11.4	School, Piau village (indig.)
AC 1139/7/67	66	78.8	10.6	Nova Floresta village (indig.)
PI 11878/28/99	268	54.5	−9.7	Rural school
SP 71072/2/519	293	55.3	−7.1	School (same site)
SP 71072/2/539	312	64.1	−7.1	School (same site)
MA 8974/70/403	127	88.2	6.9	Zemuehan Tentenhar school (indig.)
BA 34134/170/262	60	66.7	6.8	School
SP 71072/2/491	308	51.3	−6.6	School (same site)
SP 71072/2/532	313	56.2	−6.5	School (same site)
PA 4723/80/254	162	97.5	6.3	Trincheira Bacajá school (indig.)
ZZ 30341/1/3576	251	64.9	6.0	Porto, Portugal (abroad)

Turnout change The first version of the turnout test compared each section with the national median and flagged 2,088 sections, which were in fact whole states (AC, AP, RR, MA and PA) where turnout fell in the runoff. The corrected version compares each section with its own municipality. It flagged 730 sections (0.15%) with robust $|z| \geq 6$ on the log-ratio of votes between rounds: 574 had *fewer* votes in the runoff and 156 more. A keyword classification of the polling-place names (approximate) found 54 prisons or juvenile detention units, 53 of them losing voters, consistent with releases and transfers of inmates between rounds; 53 indigenous villages, 40 of them gaining voters; and 18 sections abroad. The remaining 80% are ordinary polling places. The flagged sections concentrate in PA (128), AC (85), AM (68), MA (67) and AP (64). The largest increases occurred in indigenous schools in PA and MS (e.g. PA 4111/18/519, from 21 to 35 votes; PA 5835/106/330, from 128 to 212) and in the embassy in Canberra (from 72 to 111).

6.6 Turnout–vote fingerprint (T7)

The two-dimensional histogram of turnout and leader share (Figure 5) shows a single compact cloud centred around 80% turnout, elongated along the vote axis, without the smear toward the 100%/100% corner that characterizes ballot stuffing [3, 14]. Within municipalities, turnout was *negatively* correlated with Lula's share ($r = -0.095$ in the first round and $r = -0.069$ in the runoff) and positively with Flávio Bolsonaro's ($r = 0.086$ and 0.069). Ballot stuffing in favour of the winner would produce the opposite sign. Sections with turnout $\geq 95\%$ and share $\geq 95\%$ for the same candidate are rare: 26 (first round) and 64 (runoff) for Lula and 1 and 2 for Flávio Bolsonaro, i.e. at most 0.013% of sections; they are likely the same small homogeneous sections found by T1, which remains to be confirmed. Integer turnout percentages were not in excess: 7,635 observed versus 7,581.5 expected in the first round ($p = 0.27$) and 7,088 versus 7,155.4 in the runoff ($p = 0.85$).



(a) First round, F. Bolsonaro **(b)** Runoff, Lula

Figure 5. Election fingerprints: turnout (x axis, %) versus the share of valid votes of the round leader (y axis, %); brighter means more sections.

7. Cross-analysis: Turnout × Change of Vote Profile (T8)

T8 tests whether sections whose turnout changed more than their neighbours' also changed their vote disproportionately. Inflating a section with votes for a single candidate leaves exactly that mark: more votes and a profile shifted toward the same side [3].

For each section present in both rounds with at least 30 votes in each, two measures are computed, both centred on the municipality median, using only municipalities with three or more sections:

$$\Delta C_s = \ln\left(\frac{V_{2,s}}{V_{1,s}}\right), \qquad \Delta P_s = 100\left(p_{2,s} - \hat{p}_{2,s}\right), \qquad (19)$$

where V is the section total in each round (including blank and null votes), p_2 is the share of finalist A (Lula) among the two finalists in the runoff, and $\hat{p}_2$ is the prediction of that share from the first-round composition of the same section (votes for A, for B, for other candidates, and blank or null), obtained from a transfer regression estimated by state [15]. ΔP thus measures the part of the profile change that the first round does not explain.

Deciles Figure 6 shows the mean of ΔP per decile of ΔC. The curve is not flat: it goes from −0.05 in the lowest decile (minimum −0.09 in the second) to +0.22 percentage points in the highest, so sections whose turnout grew most relative to their municipality shifted slightly toward Lula, and those whose turnout fell most shifted slightly toward Flávio Bolsonaro. The effect is statistically clear (non-overlapping 95% intervals at the extremes) but small: two tenths of a percentage point in the top decile, where turnout grew 5.1% above the municipal median.

Correlation by state The national Spearman correlation between ΔC and ΔP is $\rho = 0.045$ (0.031 without centring). It varies in sign across units (Figure 7): it is positive in 19 and negative in 9 of the 28 units, ranging from −0.10 (AP and AC) to +0.16 (sections abroad), with the largest positive

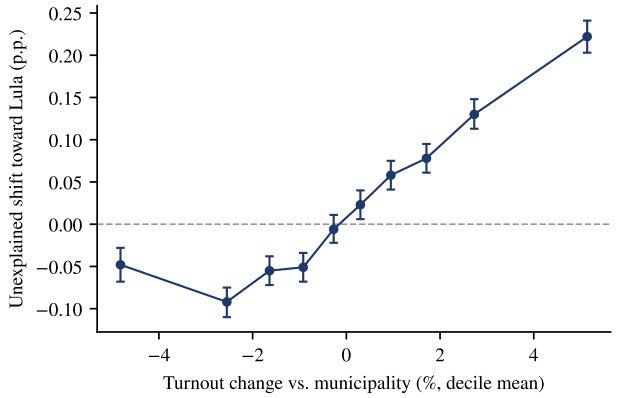


Figure 6. Mean unexplained shift toward Lula (ΔP, with 95% CI) per decile of turnout change relative to the municipality (ΔC); 498,952 sections.

values in the South and Centre-West (SC, MS, PR, DF, MT). A coordinated manipulation in favour of one candidate would push all states in the same direction; the mixed signs are more consistent with regional differences in which groups of voters were mobilized or demobilized between rounds.

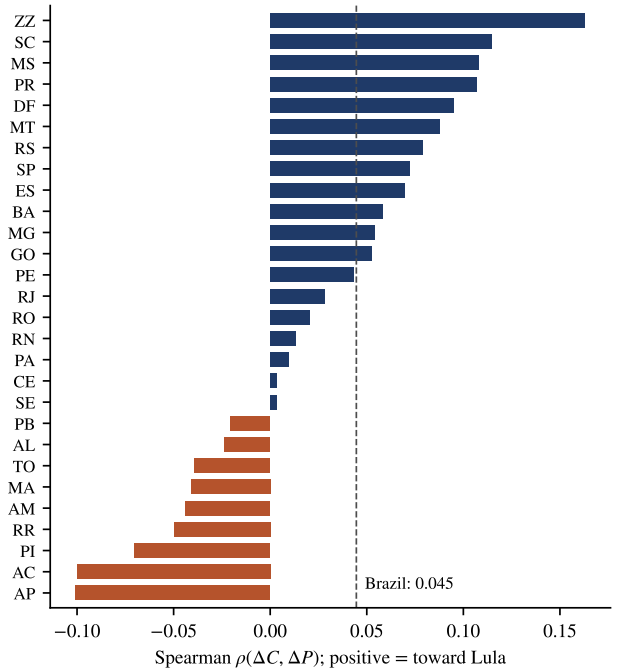


Figure 7. Spearman correlation between ΔC and ΔP by state (ZZ = abroad). Dashed line: national value.

Vote transfer The national ecological regression estimates that first-round voters of each finalist stayed with him (coefficients 1.03 and 1.01), voters of other candidates split evenly (0.44 to each), blank and null voters went mostly to Lula (0.63), and the extra runoff voters split 0.55 to Lula and 0.41 to Flávio Bolsonaro. In 9 units at least one coefficient fell outside $[0, 1]$ by more than 0.05, mostly Lula retention rates of 1.05–1.08 in SP, the South and the Centre-West. This indicates that the constant-transfer assumption of Goodman's regression does not fit there, a known limitation of ecological inference, not an impossible count.

Flagged sections and symmetry 594 sections had $|z| \geq 4$ in ΔC and $|z| \geq 2$ in ΔP (Figure 8). The pattern of ballot stuffing is "more votes, shifted toward the beneficiary". Among the 158 sections that gained voters, 88 shifted toward Lula and 70 toward Flávio Bolsonaro; the binomial test does not reject symmetry ($p = 0.18$). Among the 436 that lost voters, 263 shifted toward Lula and 173 toward Flávio Bolsonaro ($p = 1.9 \times 10^{-5}$); this asymmetry cannot come from added ballots and is consistent with the prisons, villages and urban sections found by T6. The four São Paulo sections of T6 appear again, with more votes and a shift toward Flávio Bolsonaro, which reinforces the hypothesis of a change in the electorate (transit voters) rather than an anomaly specific to one candidate.

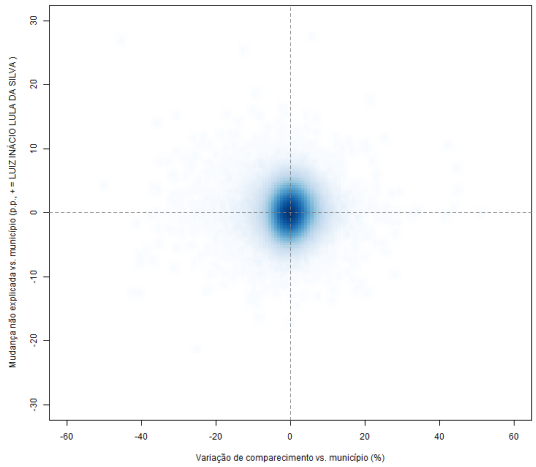


Figure 8. Unexplained shift toward Lula (y axis, p.p.) versus turnout change relative to the municipality (x axis, %); density plot.

In the validation, T8 found 4 of the 5 inflated sections, and even with the inserted frauds the overall correlation stayed at −0.005: a few inflated sections do not move the correlation. The list of sections and the symmetry test are therefore the decisive outputs: among sections that gained voters — the pattern that inflation would create — neither finalist was favoured.

8. Verification with the Voting-Machine Files (N1, N2, N4, N5)

Statistical tests can only say whether the published counts look like honest counts. The files that each voting machine writes at the end of the day allow a direct check: whether the published result is what the machine recorded, and whether the

files were altered afterwards. The TSE published these files for every section of both rounds [58]. We processed 499,207 sections per round, which covers every official section plus one or two sections abroad with files but no votes.

8.1 Recount of the digital vote record (N1)

The RDV stores each vote as cast, office by office, in random order. We decoded every RDV, recounted its votes and compared them with the BU of the same section, office by office; then we compared the BU of each section with the official per-section CSV used in Sections 6 and 7. Table 8 summarizes the result. The recount matched the BU in all 3,030,617 section–office comparisons of the two rounds: 2,490,800 in the first round (President, Senator, Governor and federal, state and district deputies, plus 10 comparisons of another contest held on the same day) and 539,817 in the runoff (President, Governor in the states with a gubernatorial runoff and 871 comparisons of municipal or supplementary contests). One section in Maranhão had no RDV file in the first round and could not be recounted. In 31 sections in the first round and 29 in the runoff the BU had been generated by the counting system rather than by the machine (29 sections abroad in each round and 2 in Minas Gerais in the first round); their RDVs also matched. The BUs, in turn, reproduced the official per-section results without a single vote of difference in any section, so the national totals are identical (Table 9).

Table 8. Recount of the RDV against the BU (N1)

	1st round	Runoff
Official sections	499,206	499,205
Sections with machine files	499,207	499,207
BU from the machine / from counting system	499,176 / 31	499,178 / 29
Sections without RDV	1	0
Section–office comparisons	2,490,800	539,817
of which divergent	0	0
Sections with BU $\neq$ official CSV	0	0

Table 9. National totals: BU files and official result (President)

Round	Option	BU files	Difference
1st	F. Bolsonaro (22)	56,104,503	0
1st	Lula (13)	53,879,538	0
1st	Others (11 candidates)	9,321,993	0
1st	Blank / null	2,300,798 / 3,669,003	0
2nd	Lula (13)	64,256,232	0
2nd	F. Bolsonaro (22)	59,826,041	0
2nd	Blank / null	1,430,496 / 2,723,305	0

8.2 Integrity and signatures (N2)

Each machine signs a file (`.vsc`) that contains the SHA-512 [37] of the BU, the RDV and the log, with a key unique to the machine and its certificate embedded. We recomputed the three hashes of every section and verified the hardware signature: ECDSA over the P-521 curve with SHA-512 [38] for models UE2013 and UE2015, and Ed521 (EdDSA over the E-521 Edwards curve [42, 38]) for models UE2020 and UE2022, the scheme the TSE reports for its newer machines [61]. All 998,414 signatures were valid and no hash differed from the signed value (Table 10); the only hashes not compared were those of the missing RDV above and of one missing log. Two limits apply. The chain from each machine certificate to the TSE root authority was not verified, because the root certificates are not published; and the second, software signature, which uses a non-public algorithm, was not verified either. The test therefore shows that the files were not altered after being signed by the key whose certificate they carry, not that this key belongs to a genuine machine.

Table 10. Hashes and hardware signatures by machine model (N2)

Model	Signature	1st round	Runoff	Failures
UE2013	ECDSA P-521	11,921	12,455	0
UE2015	ECDSA P-521	88,064	87,502	0
UE2020	Ed521	200,463	200,504	0
UE2022	Ed521	198,759	198,746	0
Total		499,207	499,207	0

8.3 Voting rhythm in the logs (N4)

The log records each event of the machine with a timestamp, including every vote cast. In the 499,176 (first round) and 499,177 (runoff) sections with a machine log, no vote was recorded before the machine was declared ready or after the session was closed; the largest number of votes in any single minute was 3 in the first round and 4 in the runoff, and no section had two votes less than 5 seconds apart. These are the marks that a burst of inserted votes would leave, and none appeared.

Figure 9 shows the share of votes per hour. In the runoff, voting was concentrated in the morning and practically ended at 17:00 (0.03% of the votes after that time); in the first round, 2.97% of the votes were cast after 17:00, by voters already in line. A likely reason is that the first round asked for six offices and the runoff for one or two, so each voter took longer and queues were longer at closing time. Within municipalities, sections with a larger share of late votes gave slightly more votes to Lula in the first round (+0.009 in the logit per percentage point of late votes; $z = 11.5$) but not in the runoff ($p = 0.13$). These late votes are in the RDV and were recounted in N1, so the association describes where queues formed, not an anomaly in the count. In 153 sections per round votes were time-stamped before 08:00 local time, but always after the machine was ready, which points to clock offsets rather than early voting. In 16 section–rounds (9 and 7) the number of vote events in the log differed from the BU; the 7 runoff cases also contained events from another date (for example, a log with 612 vote events against 308 votes in the BU in a section of Espírito Santo); in all of them the RDV recount matched the BU and the signatures were valid. They are listed for manual inspection.

8.4 Releases without biometrics (N5)

The BU reports how many voters were enabled by fingerprint and how many by biographic data (birth year) when the fin-

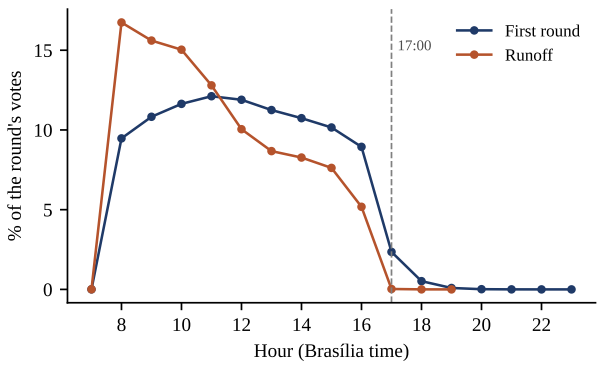


Figure 9. Share of the votes of each round cast per hour (Brasília time), from the logs of all voting machines.

gerprint was not recognized. The median share of biographic releases per section was 6.1% in both rounds (99th percentile: 19%). Using a robust z within the municipality, 589 sections in the first round and 517 in the runoff had an unusually high share ($z > 6$; 0.1% of sections). The one with the highest z, in Minas Gerais, enabled 266 of its 290 voters by biographic data, and in others the share reached 99%, which suggests a failing fingerprint reader and should be checked in the section's minutes. The share of biographic releases and Lula's vote are associated: in the top decile of biographic releases Lula received 0.59 percentage points more than the municipal mean in the first round and 0.28 in the runoff. After controlling for the share of voters with registered biometrics and for the municipality, the coefficient was small and significant only in the first round (+0.0033 in the logit per point; $p = 0.03$; runoff $p = 0.32$). Fingerprint recognition is known to fail more often for older voters and manual workers, groups that leaned toward Lula (Section 9); this hypothesis was not tested directly. Biographic releases also matter for another reason: since the poll worker can disable biometric identification, a release could be used to let someone vote in the name of an absent voter [53]. A high share of releases alone does not separate this from a failing reader; test N13 (Section 9.1) is designed to do so.

9. Further Statistical Tests (N6–N12)

Table 11 summarizes the seven further statistical tests. Unless stated otherwise, they use the runoff and the share of Lula among the two finalists; N7 and N8 use sections with at least 50 votes, N9 and N12 sections with at least 20, N12 only municipalities with at least 20 sections, and N11 sections with at least 50 eligible voters.

Machine model (N6) Without controls, sections with the oldest model (UE2013) gave Lula a larger share (51.2% of the two finalists in the first round, against 48.7–49.2% for the other models) and had lower turnout (71.6% against 78.5–79.4%). These are differences of place: older machines are used in particular regions. Within the same polling place — 7,896 places in the first round and 8,525 in the runoff had machines of two or more models — no model had a significant effect on Lula's share, turnout or blank and null votes (all BH $q \geq 0.29$; largest coefficient 0.037 in the logit). The model reported in the machine signature agreed with the model in the TSE detail file in all but 6 sections. Figure 10a shows the estimates for Lula's share: all confidence intervals of the newer models contain zero, and the larger point estimates of the oldest model (UE2013) do not survive the BH correction.

Electorate profile (N7) A logit model with municipality fixed effects relates each section's vote to the profile of its voters [56]. The coefficients have the expected signs: Lula's share grows with the shares of women, voters aged 16–24, voters with low schooling and voters with registered biometrics, and falls with the shares aged 45–59 and 60 or more (Figure 10b); low schooling has by far the largest coefficient; the profile explains 26% of the within-municipality variance of the runoff. Over-dispersion is small ($\rho = 0.013$). Using BH ($q < 0.001$), 1,024 runoff sections (0.21%) deviate from the model, about 190 of them in indigenous villages according to the polling-place name. Predictive simulation reproduces the within-municipality dispersion of Lula's first-round share ($p = 0.52$), slightly over-predicts it in the runoff (standard deviation 0.0734 observed against 0.0746 simulated) and misses the tails: there are 6,714 runoff sections with more than 90% for Lula against 4,734 simulated. The profile, which is available only by broad categories, does not capture homogeneous communities, the same limitation seen in T1.

Panel 2022–2026 (N8) 443,458 runoff sections kept the same number and polling place as in 2022 and had at least 50 votes for the two finalists in both years. Lula's share in 2026 is strongly correlated with his share in 2022 ($r = 0.973$; Figure 12), with a mean change of +0.29 percentage points and state means between −3.0 (TO) and +4.3 (DF). With a robust z centred on the municipality, 822 sections (0.19%) changed more than six robust deviations; 585 moved *away* from Lula and 237 toward him. About 90 are in indigenous villages; the most extreme is in the Maxakali indigenous land in Minas Gerais, where Lula fell from 99.5% to 70.1%. The opponent in 2026 was a different candidate from 2022, so local swings are expected. Indigenous sections are over-represented among the extreme swings, as remote areas were in Turkey [13], but here the swings mostly ran against the winner. Figure 11 shows that the mean change has different signs across states, from −3.0 points in Tocantins to +4.3 in the Federal District, and that extreme swings are concentrated in Amazonas (14 per 1,000 sections) and Maranhão (7.6), states with many indigenous villages.

Round-number spikes (N9) For each section we counted whether the share fell within 0.05 percentage points of a multiple of 5%, and compared the total with 500 binomial resamples of the same sections, which keep the spikes that small denominators create naturally. There was no excess for Flávio Bolsonaro in either round, for Lula in the runoff, or for turnout.

Table 11. Further statistical tests: method and main result

#	Test	Method	Main result
N6	Machine model	Regression with polling-place fixed effect, SE clustered by municipality, BH	No model effect on vote, turnout or blank/null within the same polling place (all $q \geq 0.29$)
N7	Electorate profile	Logit with municipality fixed effect, beta-binomial over-dispersion [19], predictive checks [29]	Expected signs; 1,024 sections flagged (0.21%); model under-predicts very high Lula shares
N8	Panel 2022→2026	Change of logit share in the same section, centred on municipality, robust z	$r = 0.973$; 822 sections with $\lvert z\rvert > 6$ (0.19%), 585 of them away from the winner
N9	Round-number spikes	Resampled kernel idea [12]: binomial resampling, 500 replicates, multiples of 5%	Only Lula, first round: +247 sections (0.05%; $p = 0.008$), spread over 30–50%
N10	Spatial clustering	Local Moran [31], 8 nearest polling places, permutations, BH	No significant cluster of extreme residuals; test under-powered (see text)
N11	Machine learning	Random forest [32, 33] trained on 300,000 synthetic sections	2 real sections flagged vs. 499 expected at 0.1% false-positive rate
N12	Local multimodality	Dip test [36] per municipality, BH; 2022 as control	8 of 3,932 municipalities in 2026 vs. 4 of 3,774 in 2022; all contain indigenous lands

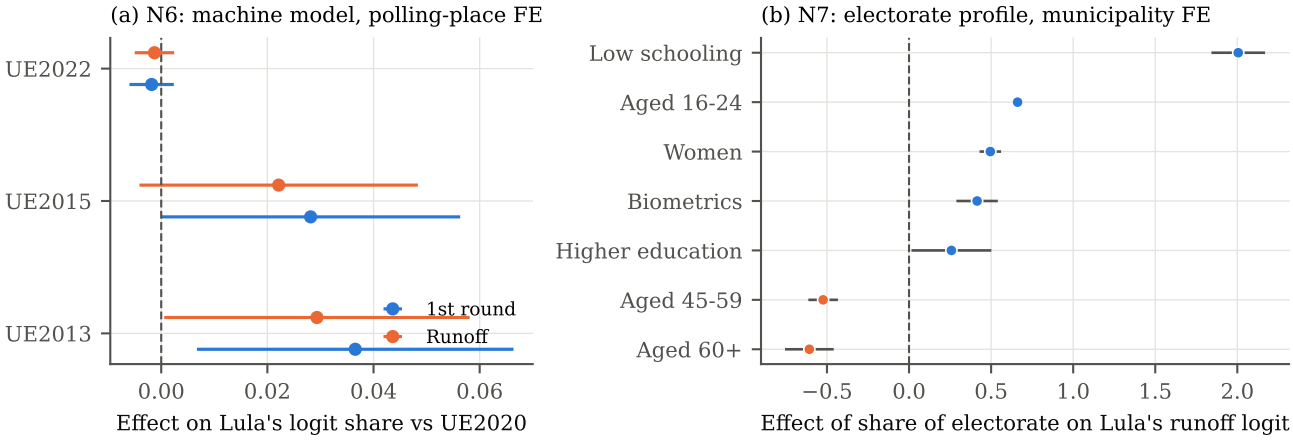


Figure 10. Fixed-effect estimates with 95% cluster-robust intervals. (a) N6: effect of the machine model on Lula's logit share within the same polling place, relative to UE2020, in each round. (b) N7: effect of the share of each group in the section's electorate on Lula's runoff logit share, within the municipality (blue: positive; orange: negative).

For Lula in the first round there were 247 more sections at round values than expected (0.05% of sections; 95% interval 0.01–0.09%; $p = 0.008$), spread over the foci between 30% and 50% rather than concentrated on high round values such as those found in falsified Russian results [11]. Given the number of series tested (six), this single small excess is weak evidence. Figure 13 shows the deviation at each round value: those of the candidates are small and of both signs, without a systematic excess at high values; turnout shows the largest deviations, of opposite signs at 80–85% and at 90%, which cancel in the total.

Spatial clustering (N10) Polling places were geocoded and linked to their eight nearest neighbours within the state. Global Moran's I was low for the residuals of the cross-round test (median across states 0.025) and of the panel (0.039), and higher for the profile model (0.200), which indicates spatially structured variables missing from that model (Figure 14); the values are highest in the Federal District (0.71) and Rio de Janeiro (0.65), large urban areas where neighbouring polling places share a social profile that the broad categories of N7 do not capture. No local cluster of extreme residuals reached significance. This null result is not informative, however: with 199 permutations the smallest attainable p-value is 0.005, which cannot survive the BH correction over about 90,000 polling places. The test must be rerun with many more permutations or with analytical inference before it can be interpreted.

Machine learning (N11) Following Zhang, Alvarez and Levin [45], we generated 150,000 clean synthetic sections, from the real first round of each section plus a typical change between rounds, and 150,000 manipulated ones, with ballot stuffing, vote swapping or fabricated round numbers, and trained a random forest of 300 trees on 11 features. On held-out synthetic data the area under the ROC curve was 0.863 (0.858 for stuffing, 0.794 for swapping and 0.938 for fabrication). With the threshold set for a false-positive rate of 0.1% on clean synthetic sections, the model would flag about 499 of the 498,950 real sections if they were all clean; it flagged 2, both in the same São Paulo polling place that appeared in T6 and T8. At this threshold, however, sensitivity is low (18% for stuffing, 15% for swapping and 0.3% for fabrication), so the test is a conservative screen. The distribution of predicted probabilities for real sections is bimodal, with a secondary mode near 0.75 holding roughly a tenth of the

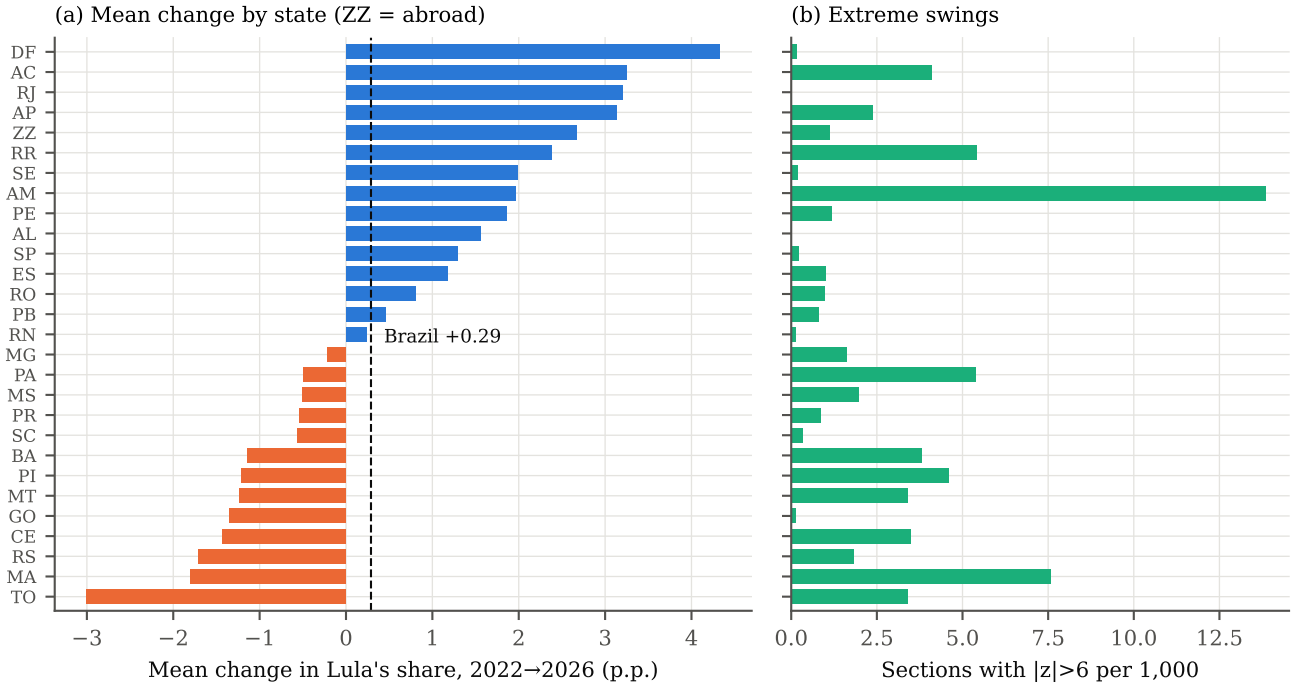


Figure 11. Panel 2022–2026 (N8) by state. (a) Mean change in Lula's share of the two finalists in the same section (dashed line: Brazil). (b) Sections with robust $|z| > 6$ per 1,000 sections of the panel.

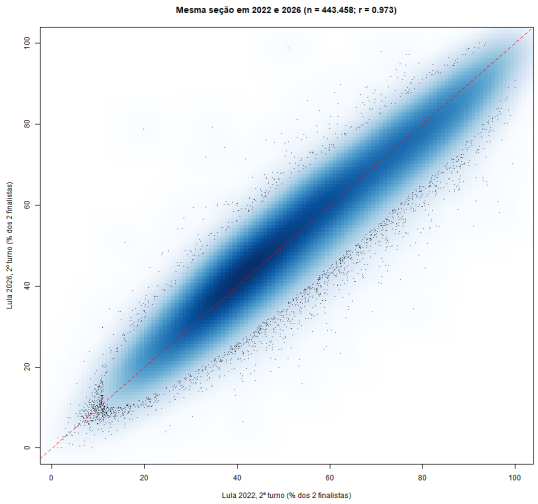


Figure 12. Same section in 2022 and 2026: Lula's share of the two finalists in the 2022 runoff (x axis) and in the 2026 runoff (y axis); $n = 443{,}458$; dashed line: no change. The dots are the sections in the least dense regions.

sections (Figure 15). The last digit of Lula's count is the second most important feature (Figure 16), and fabricated counts in the training data end in zero; since about 10% of honest counts end in zero (T2), this mode is probably an artifact of that feature, which should be confirmed by refitting the model without it.

Local multimodality (N12) A mixture of two voting regimes inside a municipality — for example, a group of inflated sections — produces a bimodal distribution of the vote share. Of the 3,932 municipalities with at least 20 sections in the 2026 runoff, 8 had a significant dip test (BH $q < 0.05$), against 4 of 3,774 in the 2022 runoff. Four are the same in both years (São Paulo de Olivença, São Gabriel da Cachoeira and Tabatinga, in Amazonas, and Japorã, in Mato Grosso do Sul); the other four are Campinápolis (MT), Tocantínia (TO), Ipuaçu (SC) and Amambai (MS), which were not multimodal, or had too few sections to be tested, in 2022. All eight contain indigenous lands, where sections in villages vote very differently from urban sections.

9.1 Biographic impersonation (N13): designed, not yet run

Impersonation of an absent voter through a biographic release would leave three marks in the *same* section: one more biographic release, one more voter and, usually, one more vote late in the day, when absences are known; a failing fingerprint reader leaves only the first. N13 uses the files already extracted for N4 and N5, without reprocessing the machine files. Its main statistic compares the two rounds of each section, so that the fixed profile of the section cancels out: with $\mathrm{d}V$ and $\mathrm{d}B$ the changes in voters and in biographic releases as percentages of the eligible voters, the slope of $\mathrm{d}V$ on $\mathrm{d}B$ with municipality fixed effects estimates the number of extra voters per extra release, close to zero if releases come from reader failures and close to one if they come from impersonation. Each

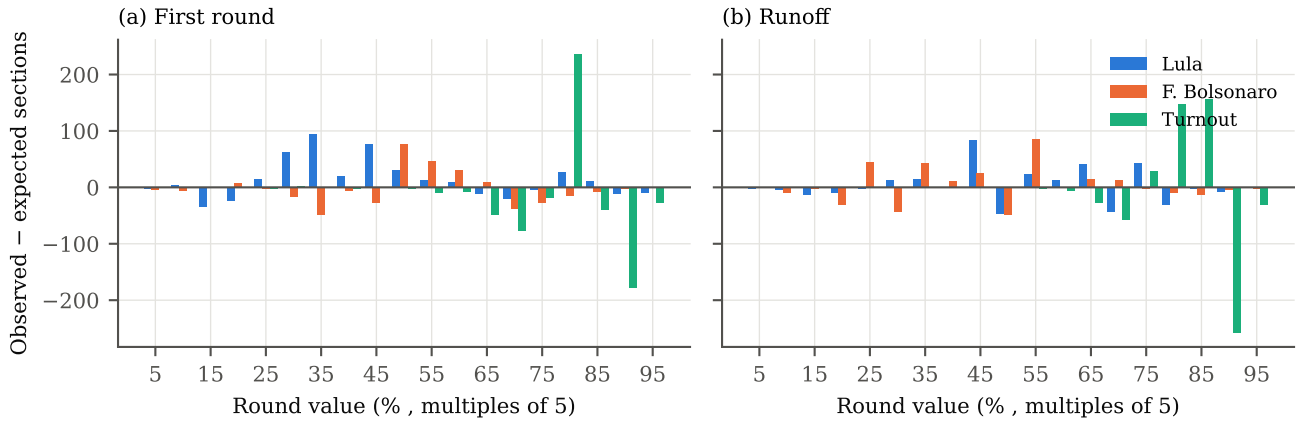


Figure 13. Round-number spikes (N9): observed minus expected number of sections within 0.05 percentage points of each multiple of 5%, where the expected value is the mean of 500 binomial resamples of the same sections.

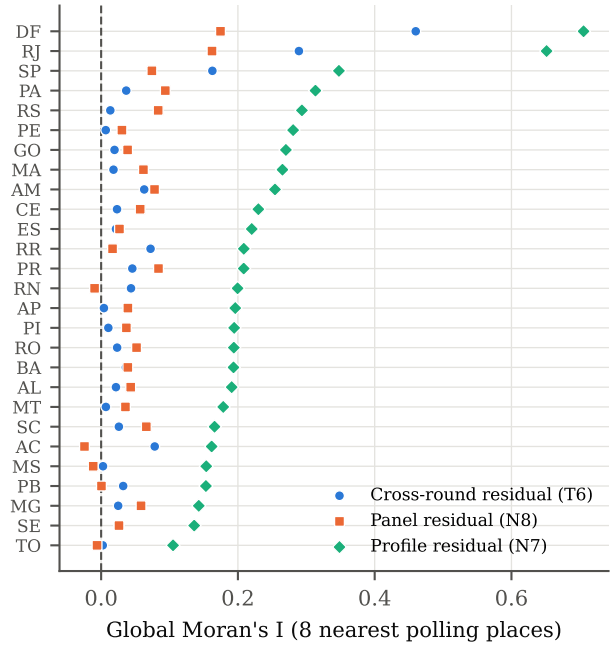


Figure 14. Global Moran's I by state (N10) for the residuals of the cross-round test, the 2022–2026 panel and the profile model; states ordered by the profile residual.

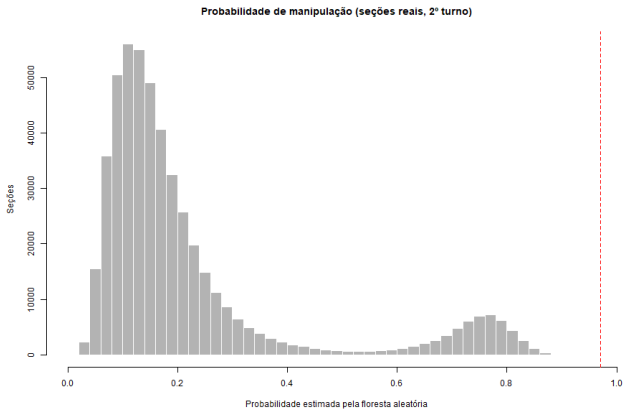


Figure 15. Probability of manipulation estimated by the random forest for the real runoff sections (x axis) and number of sections (y axis); dashed line: threshold for a 0.1% false-positive rate.

section is also scored by $S = (z_{dB} + z_{dV} + z_{dL})/\sqrt{3}$, with the robust scores (5) computed within the municipality and dL the change in the share of votes cast from 15:00 on; sections with $S > 4$ are listed and tested for symmetry between the finalists, as in T8. In a simulation with 6,000 sections that included 150 failing readers per round, the score flagged 14 clean sections (0.23%); after impersonation was inserted in 300 of them, it found 54–63 (depending on the random draw) when 10–40% of the absent voters were impersonated and 239–242 when 40–80% were, and the slope rose from 0.009 to 0.023 and to 0.08–0.09. The section score is therefore a screen for large-scale impersonation, and the slope the more sensitive aggregate test. The script (`11_teste_liberacao_biografica.R`) repeats this validation on a copy of the real data; it has not yet been run on the 2026 files.

9.2 Remaining checks

Three checks remain open. N3, the comparison of the tally sheets printed and posted at the sections with the published files, requires a stratified random sample of the QR codes collected by observers. The certificate chain of the machine signatures can only be verified if the TSE publishes the root certificates. And the classification of flagged sections by type of polling place should use the official register instead of keywords.

10. Discussion

Regarding **RQ1**, the 2026 section-level counts follow the expected regularities: digits, integer percentages, local ho-

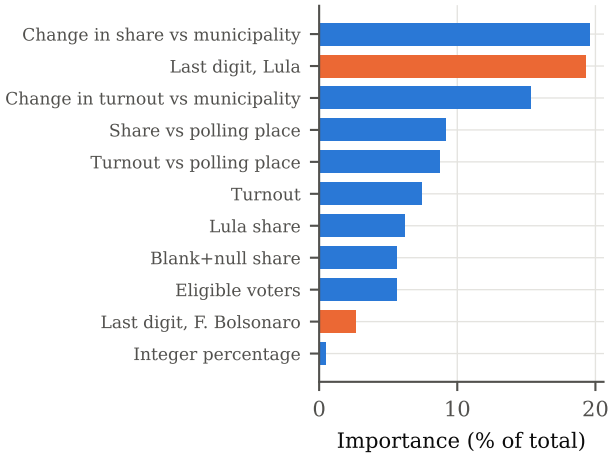


Figure 16. Importance of the features of the random forest (N11), as share of the total decrease in node impurity. Orange: last-digit features.

mogeneity and the fingerprint behave as in clean data, the runoff is predicted by the first round with $R^2 = 0.93$, the machine model has no effect within the same polling place, and a classifier trained on synthetic frauds finds fewer suspicious sections than its own false-positive rate. Regarding **RQ2**, deviations are rare (at most about 0.2% of sections, depending on the test), many repeat in both rounds, and the most extreme ones include a disproportionate number of indigenous villages, prisons and sections abroad, where the electorate is either homogeneous or changes between rounds, although most flagged sections are ordinary polling places that should be checked individually; every municipality with a bimodal vote distribution contains indigenous lands. Regarding **RQ3**, turnout change and profile change are linked by a small, broadly increasing gradient whose sign varies across states, and extreme cases that gained voters are symmetric between the finalists; among sections that *lost* voters the shift favoured Lula (263 vs. 173), a pattern that added ballots cannot produce. Regarding **RQ4**, the recount of every RDV reproduced every BU, every BU reproduced the official result, and every file carried a valid hardware signature; the logs show the rhythm of human voting.

The recount changes the meaning of the statistical results, but only up to a point. Statistical forensics can only show that the counts look honest; it cannot exclude a small, spread-out alteration that preserves every regularity. The machine files close that gap for everything that happens *after* a vote is recorded: an alteration during transmission or tallying would have to reproduce the RDV, the BU, the log and their signed hashes consistently in every affected section. They do not close it for the recording itself. The RDV and the BU are written by the same software in the same machine, so a fault or a malicious change in that software that records a vote different from the one typed would produce an RDV that agrees with the BU and files that carry valid signatures; the recount would match by construction. In the terms of Rivest [50], the Brazilian system is software dependent, and the RDV is a second electronic copy, not the voter-verified physical record that would make an independent recount possible [53]. Whether each vote was recorded as cast therefore rests on the software audits, public security tests and pre-election checks of the TSE [51, 52], not on the published data analysed here. What the statistical tests add is that, if such a fault existed, it left none of the aggregate traces that manipulation usually leaves.

These results agree with the earlier Brazilian studies for 2018 [4] and 2022 [5]. They differ from the Turkish case [13], where extreme cross-round swings in remote areas favoured the incumbent: in Brazil 2026 the remote sections with extreme changes do not favour one finalist systematically between rounds, and between 2022 and 2026 they mostly moved against the winner.

Threats to validity *Construct validity*: the statistical tests detect specific forms of manipulation — invented numbers, rounding and inflated or swapped sections — and the machine-file tests verify integrity after recording, not the correctness of recording, because the RDV and the BU come from the same software; the machine signatures were verified up to the embedded certificate but not to the root authority. *Internal validity*: binomial references assume similar voters within a polling place, which underestimates unanimity in isolated communities (T1, N7); the ecological regression of T8 does not fit in 9 units; the profile model is an approximation of the hierarchical model and does not reproduce the tails; the spatial test N10 lacked power; the classifier N11 has low sensitivity at its threshold and depends on the fraud scenarios simulated; and the classification of polling places by keywords is approximate. *External validity*: the results refer to the data and files published by the TSE up to 28 October 2026; later updates may change marginal counts. *Reliability*: all steps are scripted, seeded and saved as files, so the analysis can be reproduced by third parties.

11. Conclusion and Future Work

The 2026 Brazilian presidential result can be reproduced, vote by vote, from the files signed by the voting machines: the recount of all digital vote records matched every tally sheet, the tally sheets matched the official result exactly, and all signatures and hashes were valid. This rules out alterations after the votes were recorded; it cannot, by itself, show that the machines recorded each vote as cast, since the files are produced by the machine software. The section-level data do not show the typical patterns of count manipulation either: digit and integer-percentage tests are within the expected range; sections of the same polling place are homogeneous; the runoff of each section is predicted by its first round and by the 2022 result; the fingerprint has no smear toward full turnout and full support; the machine model has no effect; and the sections that gained voters between rounds do not favour

one finalist. The deviations found are rare; the most extreme ones are often in sections whose electorate is homogeneous or changes between rounds, and all of them are listed for individual checking.

We recommend: (i) checking the flagged sections — the 14 of T6, the 16 log–BU divergences of N4 and the most extreme biographic releases of N5 — against the minutes and tally sheets, and verifying whether the four São Paulo sections that appear in T6 and T8 (two of them also in N11) are transit-voting sections; (ii) confirming the type of the flagged polling places with the official register; (iii) rerunning N10 with adequate inference and N11 without the last-digit feature; (iv) running N3 on a field sample of printed tally sheets and asking the TSE to publish the root certificates of the machines; (v) running N13, already coded, on the extracted machine files; (vi) publishing code, data version and results openly; and (vii) communicating the findings as statistical: a flagged section is a point to check, not evidence of fraud.

Data and code availability

All data are public and were obtained from the TSE Open Data Portal [1, 58]. The scripts in Table 4 and the result files are available at ⟨https://github.com/ciuffciuff/Artigo_Eleicoes_2026⟩.

Acknowledgements

The author thanks the Graduate Program in Applied Computing of the University of Brasília.

Author contributions

The author designed the study, wrote the analysis scripts, ran the analyses and wrote the manuscript.

Declaration of AI-Assisted Use in Writing

The author used the generative AI assistant Claude (Anthropic), in October 2026, for the following tasks: translating the manuscript into English and editing it for clarity; formatting it in the journal's LaTeX template; searching the literature and checking the bibliographic data and DOIs of the references; helping to write and debug the R and Python analysis scripts; and writing the plotting code for Figures 6, 7, 9, 10, 11, 13, 14 and 16, which draws the charts deterministically from the result files produced by the scripts. No research data, result, image or chart was generated or altered by AI: all data come from the TSE Open Data Portal, all analyses were run by the author with the published scripts, and every figure can be regenerated from the repository. The author reviewed all AI-assisted text, code and references, checked every number in the manuscript against the result files, and takes full responsibility for the content of the article.

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